PRENATAL MATERNAL SMOKING AND OFFSPRING BIRTH WEIGHT: A SIBLING FIXED-EFFECTS ANALYSIS USING THE NATIONAL LONGITUDINAL SURVEY OF YOUTH 1979 CHILD AND YOUNG ADULT COHORT

Tauras Lu,

Bachler of Science in Economics, University of Nottingham Ningbo China, 2024

Thesis Advisor: Raymond Niaura, PhD.

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Department of Biostatistics

School of Global Public Health, New York University

Introduction

Low birth weight (LBW, <2,500g) affects approximately 14.7% of births globally and is associated with adverse developmental, cognitive, and economic outcomes across the life course. Maternal prenatal smoking is among the most significant modifiable risk factors for LBW, with a pooled odds ratio of 1.89 across 55 cohort studies. Despite declining prevalence, approximately 5% of U.S. births in 2020 were still exposed to maternal smoking during pregnancy. While prior research has documented associations between prenatal smoking and birth outcomes, few studies have accounted for confounding by shared familial factors using within-family designs. Unmeasured genetic and environmental factors are likely to co-occur with maternal smoking behavior; a sibling-based approach is therefore needed to obtain less confounded estimates of the effect of prenatal smoking exposure on offspring birth weight.

Methods

A retrospective cohort study was conducted using data from 8,744 children drawn from the National Longitudinal Survey of Youth 1979 Child and Young Adult Cohort (NLSY79 CYA). The primary exposure was maternal prenatal smoking (yes/no) and the outcome was offspring birth weight in ounces. Three sequential models were estimated: an unadjusted linear regression, a multivariable-adjusted linear regression controlling for child sex, race, gestational age, maternal AFQT score, and maternal age, and a sibling fixed-effects model to account for time-invariant shared family-level confounding. Mother fixed effects were defined by 2,775 unique mothers across 7,093 observations.

Results

In the unadjusted model, prenatal smoking was associated with a mean reduction in birth weight of 6.53 ounces ($p<0.001$). After adjusting for sociodemographic covariates, the estimate attenuated modestly to −6.03 ounces ($p<0.001$). In the sibling fixed-effects model, the estimated effect further attenuated to −2.97 ounces (95% CI: −4.92, −1.01; $p=0.003$), representing approximately 55% attenuation from the unadjusted estimate, suggesting substantial confounding by shared family-level factors.

Conclusion

Results from this study suggest that while prenatal maternal smoking is independently associated with reduced offspring birth weight, a meaningful portion of the crude association is attributable to unmeasured familial confounding. Policies targeting smoking cessation during pregnancy should be considered alongside broader family-level socioeconomic interventions. However, the residual effect

observed in the sibling fixed-effects model indicates that prenatal smoking retains an independent biological effect on fetal growth, consistent with nicotine-mediated placental dysfunction and carbon monoxide-induced fetal hypoxia.

Key Words: Prenatal Smoking; Birth Weight; Sibling Fixed-Effects; NLSY79; Familial Confounding

Introduction

Birth weight is among the most reliable early predictors of long-term developmental and economic outcomes. When an infant is born weighing less than 2,500 grams, they face immediate risks in the neonatal ward and potential hurdles that stretch deep into adulthood.[1] Research consistently links low birth weight to cognitive delays,[2] lower grades in school,[3] and even reduced earnings in the labor

market.[4] Because these effects are so persistent, identifying modifiable factors during pregnancy remains a priority for maternal and health research.

Maternal prenatal smoking is among the most well-established modifiable risk factors. Evidence from a recent meta analysis of 55 cohort studies indicates that smoking significantly increases the odds of low birth weight, with an estimated pooled odds ratio of 1.89.[5] Even though fewer people smoke today than thirty years ago, about 5% of American births in 2020 were still exposed to prenatal tobacco.[6] This burden is not shared equally, as it falls most heavily on younger women and those facing economic hardship. Clinical groups like the American College of Obstetricians and Gynecologists emphasize that quitting is one of the single most effective ways to improve birth outcomes.[7]

However, a persistent methodological hurdle pervades this literature: Smoking is not a random choice. Mothers who smoke often face a overlapping challenges, ranging from genetic predispositions to high-stress living conditions, that independently hinder fetal development. [8] While standard statistical models try to adjust for these variables, they frequently miss the nuances of unmeasured familial factors.[9,10] Consequently, what we interpret as a direct causal link between smoking and birth weight may actually be a reflection of these hidden confounders.

To navigate this, this study utilized a sibling fixed-effects design. By comparing two children born to the same mother under different smoking conditions, we can effectively "neutralize" shared family traits like genetics or long-term health habits.[11] Although this method has been applied to areas like substance abuse or physical injury, its use in examining birth weight within US longitudinal cohorts is surprisingly sparse. [8,9,12] Using longitudinal NLSY79 data, three sequential models were estimated to examine whether the association between smoking and birth weight persists after accounting for within-family comparisons.

Literature Review

Epidemiological Evidence

The link between smoking during pregnancy and lower birth weight is among the most robust findings in perinatal epidemiology. One comprehensive meta analysis that looked at 55 cohort studies over several decades found that prenatal smoking nearly doubled the odds of a baby being born at a low birth weight.[5] The data demonstrated a clear dose response gradient, with greater daily cigarette

consumption associated with larger reductions in birth weight. These results are consistent with earlier evidence from across the Americas, reinforcing that this is not just a localized issue but a robust global pattern.[13]

The effect is not limited to the birth weight change. Other ways to measure how a foetus grows also show that smoking during pregnancy is bad for it. When a mother smokes during the early stages of her pregnancy, there is a higher risk of preterm birth and small-for-gestational-age status.[14] Some studies even show long term physical effects, such as smaller head circumferences or different growth patterns that persist into childhood.[15] To address concerns about self-report validity, Knopik et al. used genetic data and biological samples to confirm the results. Their work showed that the effect of smoking on birth weight is detectable across multiple measurement approaches.[16] Other research using genetic instrumental variables even suggested that the the actual causal effect may exceed the estimates provided by standard regression models.[17]

The Role of Familial Confounding

Despite the consistency of these findings, a methodological worry remains: the potential for unmeasured familial confounding. Mothers who smoke during pregnancy are systematically different from non-smokers along dimensions that are difficult to fully capture. These differences encompass genetic predispositions to addiction, chronic stress, dietary influences, and economic stability, and are challenging to quantify in survey data. If these same factors are what actually drive restricted fetal growth, then standard regression models likely overstate the direct impact of tobacco on birth weight.

This concern is not merely theoretical. D'Onofrio et al. demonstrated that the observed association between maternal smoking during pregnancy and offspring substance use problems nearly vanised once they used a quasi-experimental design to account for shared family traits. This suggests that much of the "raw" association is driven by selection bias rather than a biological intrauterine effect. [8] Similar patterns have been observed for behavioral outcomes; when propensity score matching is used to adjust for maternal characteristics, the association between prenatal smoking and offspring externalizing problems was considerably reduced compared to unadjusted estimates, This highlights the inadequacy of conventional covariate adjustment when unmeasured confounders are present.[9] Evidence from longitudinal studies further complicates the picture, suggesting that prenatal tobacco smoke exposure is correlated with postnatal environmental factors that also influence neurobehavioral development,

making it difficult to peel the intrauterine effect away from the broader family environment in which children are raised.[18]

The confounding challenge is further compounded by the correlation between maternal cognitive ability and smoking behavior. Research using the Armed Forces Qualifications Test score as a proxy for maternal pre-market cognitive ability and human capital has consistently shown that cognitive ability is associated with both smoking propensity and offspring developmental outcomes, underscoring the importance of controlling for maternal cognitive resources when estimating the effect of prenatal smoking.[19] Failing to account for this dimension of maternal capacity risks confounding the smoking effect with the broader influence of maternal human capital on birth outcomes.

Sibling Based Designs as a Methodological Response

The sibling fixed effects design has emerged as an important quasi-experimental tool for addressing unmeasured familial confounding in observational studies. By restricting comparisons to siblings raised within the same family, the design implicitly adjusts for all characteristics that are constant across births to the same mother. These include genetic background, household socioeconomic status, neighborhood environment, and stable parental behaviors, regardless of whether those characteristics are actually measured in the data.[11] This approach is especially valuable for studying prenatal exposures, where the many factors that occur alongside smoking are both wide ranging and difficult to track individually.

A key methodological consideration in these designs is the difference between confounders shared within families and those that change between births. Frisell and colleagues demonstrated that sibling comparison estimates can still be biased if non shared confounders are not handled correctly.[20] Factors like birth order, maternal age at delivery, or changing household conditions can differ between siblings and still lead to incorrect results. This implies that sibling fixed effects estimates should be viewed as an upper bound on how much family level confounding explains the initial association, rather than as a perfectly causal estimate on its own.

Applied studies using sibling designs have consistently shown that familial confounding is a source of bias in prenatal smoking studies. A quasi experimental sibling analysis using Swedish population data found that the effect of maternal smoking on birth weight was meaningfully smaller within families than in the general population.[21] These patterns suggest that shared family factors account for a large

share of the original association. More broadly, a meta analysis comparing full cohort results to sibling comparison results across multiple health outcomes found that sibling based estimates were consistently attenuated toward the null.[22] The authors attributed this pattern partly to the control of unmeasured family confounders and partly to the unique statistical properties of within family designs. Together, these findings motivate the current study, which applies a sibling fixed effects design to a United States based longitudinal cohort to quantify the contribution of familial confounding between prenatal smoking and birth weight.

Methods

Data Source and Study Population

This study used data from the National Longitudinal Survey of Youth 1979 Child and Young Adult Cohort (NLSY79 CYA), a nationally representative longitudinal dataset that follows children born to female respondents of the NLSY79 from 1986 onward.[23] The NLSY79 is sponsored by the U.S. Bureau of Labor Statistics and has been administered biennially since 1994, with high response rates sustained across survey rounds.[23] Because the NLSY79 oversampled Black, Hispanic, and economically disadvantaged populations, it provides particular analytic value for studying the relationship between socioeconomic exposures and child health outcomes in demographically diverse samples.

The analytic sample consisted of 8,744 children with non-missing data on both smoking during pregnancy and the baby's birth weight. Children with missing values on exposure or outcome were excluded. The sibling fixed-effects model was further restricted to 7,093 children born to 2,775 unique mothers. Each of these mothers had at least two children in the survey. Within-family singleton observations contribute no family-level variation and are therefore uninformative for fixed-effects estimation.

Exposure, Outcome, and Covariates

The primary exposure of interest in this study was maternal prenatal smoking. This was treated as a binary variable based on maternal reports during each specific pregnancy. The outcome was measured as the child's birth weight in ounces, derived from birth records and maternal reports.

Covariates included in the multivariable-adjusted model were selected based on a priori theoretical considerations regarding their role as potential confounders of the smoking-birth weight relationship.[24] Child-level covariates included child sex (male/female), child race and ethnicity (Hispanic, Black, Non-Black Non-Hispanic), and gestational age in weeks, which is a strong determinant of birth weight independent of smoking exposure.[25] Maternal covariates included age at delivery and Armed Forces Qualifications Test (AFQT) score, a composite measure of verbal and mathematical ability administered to NLSY79 respondents in 1980 and widely used as a proxy for maternal pre-market cognitive ability and human capital.[19] Prenatal alcohol use was also included as a binary covariate given its established association with fetal growth restriction. Birth year of the child was included in the sibling fixed-effects model to account for secular trends in birth outcomes that may vary across pregnancies to the same mother.

Statistical Analysis

The association between prenatal smoking and offspring birth weight was assessed through three sequential regression models. This approach allows for a clear quantification of how shared familial confounding contributes to the observed results.

Model 1 consisted of a basic ordinary least squares regression with no adjustments. This model provides the crude estimate between smoking exposure and birth weight, serving as a baseline for the subsequent analyses.

$$BW_i = \beta_0 + \beta_1 Smoke_i + \varepsilon_i$$

Model 2 expanded upon the initial model by incorporating the full set of child and maternal covariates. By including factors such as gestational age, maternal age, and cognitive ability, this model reflects the standard approach in observational research.[26] It provides an estimate of the association after accounting for the measured confounders available in the survey data.

$$BW_i = \beta_0 + \beta_1 Smoke_i + \beta_2 Sex + \beta_3 Race_i + \beta_4 GestAge_i + \beta_5 Alcohol_i + \beta_6 AFQT_i + \beta_7 MomAge_i + \varepsilon_i$$

Model 3 utilized a sibling fixed effects design to isolate within family variation in smoking exposure across siblings born to the same mother. By including a fixed effect for each mother, this model automatically adjusts for all time invariant family characteristics. This includes maternal genetics, long

term health habits, and the stable home environment, even if those factors were not explicitly measured.[11] In this framework, each mother effectively serves as her own control. The model also accounted for non shared confounders that might change between births, such as child sex and birth year.[20]

$$BW_{ij} = \beta_1 Smoke_{ij} + \beta_2 Sex_{ij} + \beta_3 Race_{ij} + \beta_4 BirthYear_{ij} + \alpha_j + \varepsilon_{ij}$$

The primary focus of this analysis was the degree of attenuation observed from Model 1 to Model 3. A significant reduction in the size of the smoking coefficient would suggest that the initial association was driven largely by shared family factors rather than a direct intrauterine effect. All statistical work was performed in R using the fixest package for the fixed effects estimations. Since the study relied on public data, it was exempt from formal institutional review.[24]

Considering moderate to large standardized mean differences across several variables, a propensity score matching analysis was performed to test the strength of the multivariable results. This process involved using nearest neighbor matching on the logit of the propensity score with a 1:1 ratio. This matching was performed without replacement and utilized the exact same set of variables as the previous model.[27] To confirm that this process effectively improve the covariates balance between groups, the balance between the groups was checked using standardized mean differences. Any values falling below 0.10 were considered indicative of adequate balance for a fair comparison.[28]

Results

Descriptive Statistics

The analytic sample included 8,744 children born to 4,426 unique mothers. Within this group, 2,915 children (33.3%) were exposed to maternal prenatal smoking. Mothers who smoked tended to be younger at delivery, had lower scores on the Armed Forces Qualifications Test, and were more likely to report using alcohol during pregnancy compared to those who did not smoke. There were also significant differences in race and ethnicity between the groups, with children identified as Non Black Non Hispanic being overrepresented in the smoking group. The average birth weight was 118.71 ounces for children of non smoking mothers and 112.18 ounces for children of smoking mothers. While birth weight differed, gestational age remained similar across both groups. Detailed descriptive statistics are provided in Table 1.

The sibling fixed effects sample was narrowed to 7,093 children nested within 2,775 mothers. Among these families, 454 were exposure discordant, meaning the mother smoked during at least one pregnancy but remained smoke free during another. These discordant families provide the identifying variation for the within-family estimates

Regression Results

The results from all three models are summarized in Table 2. In the unadjusted model, prenatal smoking was associated with a mean reduction in birth weight of 6.53 ounces. After adjusting for child sex, race, gestational age, alcohol use, maternal test scores, and age at delivery in Model 2, this estimate declined slightly to 6.03 ounces. This modest attenuation suggests that observed sociodemographic characteristics account for a relatively small share of the crude association.

However, the sibling fixed effects model present a substantially greater attenuation. In Model 3, the estimated effect of smoking on birth weight fell to 2.97 ounces. This represents a 55% reduction from the original unadjusted estimate. Such a large reduction indicates that a substantial part of the link seen in the general population is actually driven by shared family factors rather than a direct effect of smoking on the fetus.

Sensitivity Analysis

Prior to matching, there were clear imbalances between the smoking and non smoking groups. The standardized mean differences ranged from 0.27 to 0.45 across several key variables, including race, alcohol use, test scores, and maternal age. These figures indicate that the two groups differed in ways that could easily skew the results if not properly handled. After applying a 1:1 nearest neighbor propensity score match, these differences were largely eliminated. In this matched sample, the differences between groups dropped to a much more acceptable range of 0.04 to 0.10.

The regression estimate in this new balanced sample was 5.71 ounces, which is remarkably close to the 6.03 ounce estimate found in the multivariable adjusted model. Because these two fundamentally different methods of controlling for variables produced nearly identical results, there is greater confidence that the findings from Model 2 are not an artifact of one specific adjustment strategy.

Instead, the results appear to be a robust reflection of the data, holding up even after correcting for the initial imbalances.

Discussion

Principal Findings

This study estimated the association between maternal smoking during pregnancy and offspring birth weight by using a sibling fixed effects design within a nationally representative group of American families. Initially, the data showed a substantial association of approximately 6.53 ounces; however, this figure was slashed by more than half once accounting for shared familial confounders. Even after this major reduction, a within family estimate of 2.97 ounces remained statistically significant. These results suggest that while smoking during pregnancy retains an independent and harmful effect on fetal growth, traditional observational studies likely inflate this impact because they fail to account for unmeasured family confounding.

Comparison with Prior Literature

The within family estimate of roughly 3.0 ounces is broadly consistent with other quasi experimental research. For instance, a sibling study using Swedish register data also found that the effect of smoking on birth weight was much smaller within families than in the general population.[21] This pattern suggests that a large portion of the original link is attributable to shared family-level characteristics rather than the smoking itself. Similar findings appear in genetic studies, which consistently show smaller effects once family genetics and environments are controlled for.[16] While some genetic instrumental variable models suggest the true causal effect could be larger, those results depend heavily on how the genetic markers are used.[17] In general, the tendency for sibling based estimates to be attenuated to null is well documented across many different health outcomes.[22] It is consistent wtih the theory that smoking during pregnancy is deeply tied to other family-level factors that also influence health.[8]

The remaining 3.0 ounce effect, while smaller than unadjusted numbers, is still clinically important. Even a modest drop in birth weight can lead to real risks later in life, such as slower cognitive development,[2,3] and even reduced earnings as an adult.[4] Some estimates suggest that a one pound loss in birth weight can lead to a 7% drop in adult wages, showing that what happens in the womb has long

term economic consequences.[29] Because this effect stays significant even after controlling for family background, it represents a real harm that cannot be attributed solely to familial selection.

The impact of prenatal smoking also goes beyond birth weight. Data from other cohort studies show that children exposed to smoking often face height deficits and higher rates of obesity by age ten.[30] These findings show that the damage caused by smoking has lasting effects on a child's growth that persist well beyond the neonatal period.

Biological Mechanisms

The fact that a within family effect persists is consistent with established biological evidence. Nicotine causes vasoconstriction in placental vessels, which impairs uteroplacental blood flow and reduces oxygen and nutrient delivery to the developing fetus.[31] Carbon monoxide from cigarette smoke also binds to fetal hemoglobin with higher affinity than oxygen.[32] This raises the levels of carboxyhemoglobin in the fetus and impairs fetal oxygen transport. Together, these biological processes provide a plausible pathway for how smoking directly constrains fetal growth regardless of the mother's social or economic background.

Limitations

It is important to acknowledge the boundaries of this study. First, a sibling fixed-effects model only adjusts for factors that remain static across all of a mother's pregnancies. Time-sensitive variables, such as fluctuations in household income, could still skew results if not fully captured. [20] Second, the analytical power of this study rests on a specific subset: the 16.4% of families where a mother smoked during one pregnancy but not another. This smaller sample naturally leads to less precise estimates than a broader cohort might offer. [22] Furthermore, smoking was measured as a binary indicator, which glosses over variations in frequency or duration. Because heavy smokers likely experience more pronounced effects, this binary approach may miss critical nuances. [33] Additionally, because the data relied on mothers reporting their own smoking habits, some may have underreported due to the stigma involved. In sibling studies, this kind of misclassification usually makes the effects smaller than they actually are.[20] Finally, since the NLSY79 data spans several decades, it is worth questioning whether

these same patterns of familial confounding hold true in more modern social and clinical environments.[34,35]

Policy Implications

Despite limitations, the findings carry clear public health takeaways. The fact that a statistically significant effect remains after controlling for family background is vital. It reinforces the need for robust smoking cessation programs for pregnant women. Official guidelines already advocate counseling and medical support as a standard part of prenatal care,[7] and this study provides new evidence to support those recommendations. However, the dramatic reduction in the effect size when looking within families suggests that smoking is not an isolated behavior; it is deeply embedded in a mother's socioeconomic reality. Since smoking is now increasingly concentrated among younger and economically marginalized women, more integrated approaches that combine quitting support with broader family-level socioeconomic interventions are required.[6,36]

Conclusion

By applying a sibling fixed-effects design to the NLSY79 Child and Young Adult Cohort, this study sought to clarify the relationship between maternal smoking and infant birth weight. By accounting for unmeasured family confounding across three sequential models, the data showed that the estimated effect dropped from initial 6.53 ounces to 2.97 ounces in the within family model, a 55% reduction. This shift suggests that a major portion of the association observed in general populations is actually a byproduct of shared genetic and environmental factors. Nevertheless, the stubborn persistence of a significant within-family effect confirms that prenatal tobacco exposure has an independent biological cost that cannot be explained away by familial selection alone.

These findings fit within the broader framework of the developmental origins of health and disease. In this view, intrauterine exposures shape physiological trajectories that extend well beyond birth weight itself.[37] Even modest reductions in weight at birth are linked to slower cognitive development,[3] diminished adult earnings,[4] and elevated chronic disease risks throughout life.[2] Therefore, the within family effect documented here carries real public health significance that persists even after the most rigorous adjustments available in observational data.

From a policy standpoint, these results support continued investment in smoking cessation programs targeting pregnant women. Evidence from large-scale randomized trials demonstrates that psychosocial support during pregnancy reduces the number of low birth weight babies and improves health outcomes for newborns.[38] This provides a practical way to prevent the biological harm identified in this study. However, the large attenuation in the effect size across the models also highlights that smoking during pregnancy is deeply tied to socioeconomic disadvantage. Smoking is one of the most important mediating factors linking low social status to restricted fetal growth.[39] Because of this, cessation programs will likely have the greatest impact when they are part of broader efforts to improve the social and material conditions of the most socioeconomically disadvantaged women.[40] Future research should examine if the level of family confounding has changed in more recent birth cohorts, as the socioeconomic patterns of prenatal smoking continue to evolve.

Appendix 1: Foundational Competencies and Concentration-specific Competencies

FC1: Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming and software, as appropriate.

FC2: Interpret results of data analysis for public health research, policy or practice.

FC3: Apply epidemiological methods to settings and situations in public health practice.

CC1: Translate hypotheses and associated outcome measures into appropriate study designs.

CC2: Implement statistical analysis plans based on the hypotheses, study design, and data features.

Table 1: Sociodemographic Characteristics of Study Participants, NLSY79 Child and Young Adult Cohort (N = 8,744)

Variable Name	Non-Smoking Mothers (N = 5829)	Smoking Mothers (N = 2915)	SMD
Outcome			
Birth Weight (oz), mean (SD)***	118.71 (21.01)	112.18 (21.81)	0.305
Exposure			
Maternal Smoking During Pregnancy, n (%)	5829 (100.0)	2915 (100.0)	-
Covariates			
Child Sex, n (%)			0.02
Male	2968 (50.9)	1514 (51.9)	
Female	2861 (49.1)	1401 (48.1)	
Child Race/Ethnicity, n (%)***			0.356
Hispanic	1283 (22.0)	309 (10.6)	
Black	1575 (27.0)	685 (23.5)	
Non-Black, Non-Hispanic	2971 (51.0)	1921 (65.9)	
Birth Year, mean (SD)***	1986.29 (5.91)	1984.44 (5.43)	0.326
Alcohol Use During Pregnancy			0.45

n (%)***			
No	3679 (63.1)	1201 (41.2)	
Yes	2150 (36.9)	1714 (58.8)	
AFQT Score, mean (SD)***	40.83 (28.83)	33.66 (24.52)	0.268
Maternal Age at Birth, mean (SD)***	25.84 (5.54)	24.27 (5.13)	0.295

**** p < 0.001. SMD = standardized mean difference. Significance tests based on chi-square for categorical variables and t-test for continuous variables.*

Table2: Association Between Maternal Prenatal Smoking and Offspring Birth Weight Across Three Sequential Models, NLSY79 Child and Young Adult Cohort

Model	Covariates	N	Estimate (oz)	95% CI
Model 1: Unadjusted	None	8,744	−6.526***	(−7.472, −5.579)
Model 2: Multivariable Adjusted	Child sex, race, gestational age, alcohol use, AFQT score, maternal age	8,744	−6.029***	(−6.883, −5.175)
Model 3: Sibling Fixed-Effects	Child sex + birth year + mother fixed effects	7,093	−2.965***	(−4.915, −1.016)
Sensitivity: PSM	Same as Model 2, matched sample	5,830	−5.705***	—

*PSM = propensity score matched sample (1:1 nearest-neighbor matching). Post-match SMD range: 0.004–0.098. *** p < 0.001.*

Table3: Family Structure and Within-Family Smoking Variation (N = 2,775 Families)

Family Size	N Families (%)	N Discordant Families (%)
2 children	1662 (59.9)	208 (12.5)
3 children	781 (28.1)	157 (20.1)
4 children	258 (9.3)	70 (27.1)
5+ children	74 (2.7)	19 (25.7)
Total	2775 (100.0)	454 (16.4)

Discordant families are defined as families in which maternal prenatal smoking status varied across siblings. Only discordant families contribute identifying variation to the sibling fixed-effects estimator.

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